

Spatiotemporal Graph Neural Networks

Graph Deep Learning, Spring Semester 2022



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March 14, 2022

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What we are going to see in this lecture

1. **Graphs and time dimension**

- Examples
- Taxonomy of temporal graph signals

2. **Spatiotemporal graph signals**

- Sensor Networks
- Graph sequences

3. **Spatiotemporal GNNs**

- Time-then-Space approach
- Time-and-Space approach
- Spatiotemporal Message Passing

4. **Spatiotemporal imputation**

- Missing data imputation
- Virtual sensing

5. **Coding Spatiotemporal GNNs**

- ts1: a PyTorch library for spatiotemporal data processing

Graphs and the time dimension

We saw how attributed graphs are effective in modeling **relational information** and in accounting for the **structure** of many physical systems.

However:

- Interactions might happen **over time**.
- (Cyber-)physical systems are often made of **sensor networks** that acquire **spatiotemporal** data streams.

Examples



- Interaction Networks
 - Social Networks
 - Recommender Systems

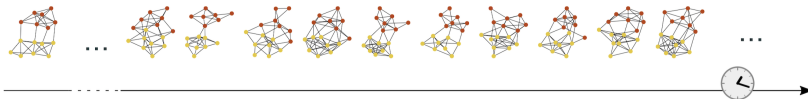


- Heterogenous wireless sensor networks
 - Traffic Networks
 - Smart grids

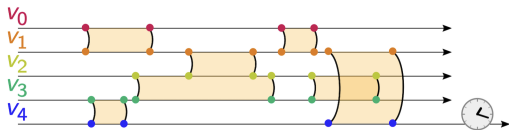
How to model such systems?

Taxonomy of temporal graph signals (i)

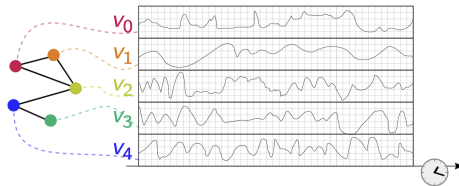
There are several settings in which time comes into play when considering graph data.



Graph streams



Dynamic Graphs

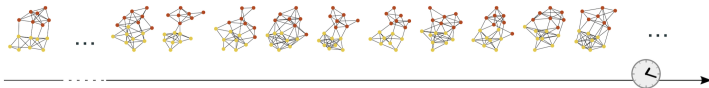


Spatiotemporal graph signals

Image credits: Daniele Zambon.

Graph streams

We start from the most general setting.



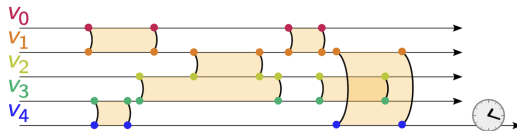
- Graphs are sampled from a **stochastic process** $\mathcal{G}_t \sim P$.
- Nodes are **not identified**. $\rightarrow$ No correspondence between nodes at different time steps.
- Arbitrary **changes** in topology.
- Difficult to track changes, **defining statistics is not trivial**.
- A good reference is Daniele's PhD thesis (see iCorsi).

[1] Zambon et al., "Concept drift and anomaly detection in graph streams", TNNLS 2018

[2] Zambon et al., "Change-point methods on a sequence of graphs", TSP 2019

Dynamic graphs

We can look at interactions that happen over time as sequences of **relational events**.



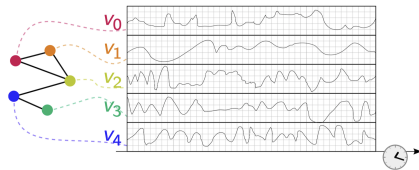
- Dynamic graphs model systems where **relationships** and node attributes **evolve over time**.
- Event-based paradigm: dynamic graphs can be seen as a **sequence of interactions** among nodes.
- Powerful paradigm to model **social/interaction networks** and build **recommender systems**.
- On iCorsi you'll find the presentation on this topic given by **Emanuele Rossi** (Twitter) in the previous edition of this course.

[3] Rossi et al., "Temporal Graph Networks for Deep Learning on Dynamic Graphs", 2020

Spatiotemporal graph signals

Spatiotemporal graphs capture the setting typical of **sensor networks**.

- A different approach to model **multivariate time series** coming from multiple sources.
- Each node (**sensor**) is associated with a time series (eventually with multiple channels)
- Edges describe (soft) **functional dependencies** among sensors
 - E.g.: causality, physical constraints, etc.



We will focus on these settings.

Spatiotemporal graph signals

We refer to *Sensor Networks* as systems where

- A **set of nodes** (sensors) observe a set of values at each time step.
- Each node is **identified** (allowing us to talk about time series).
- Nodes constitute what we refer to as the **spatial** dimension.
- Examples: traffic networks, air quality monitoring stations, smart grids, hydrological networks.

We model time series coming from sensor networks as **sequences of graphs**.

[4] Cini et al., "Neural spatiotemporal forecasting" (in preparation)

Spatiotemporal graph signals (i)

We consider a **graph signal** as a tuple $\mathcal{G}_t = \langle \mathbf{X}_t, \mathbf{U}_t, \tilde{\mathbf{A}}_t, \mathbf{E}_t \rangle$ where

- $\mathbf{X}_t \in \mathbb{R}^{N_t \times d_o}$ is the **node-attribute matrix**
 - $\mathbf{x}_t^i$ is the i -th row of $\mathbf{X}_t$ and indicates the d -dimensional node attribute vector associated with the i -th node
- $\mathbf{U}_t \in \mathbb{R}^{N_t \times d_u}$ are **exogenous variables** $\mathbf{u}_t^i$ (e.g., weather forecasts, datetime information)
- $\mathbf{E}_t \in \mathbb{R}^{N_t \times N_t \times d_e}$ is an (often sparse) **edge attribute matrix**.
- $\tilde{\mathbf{A}}_t \in \mathbb{R}^{N_t \times N_t}$ is a **weighted adjacency matrix**.

We focus on settings where the number of nodes is fixed, i.e., $N_t = N$.

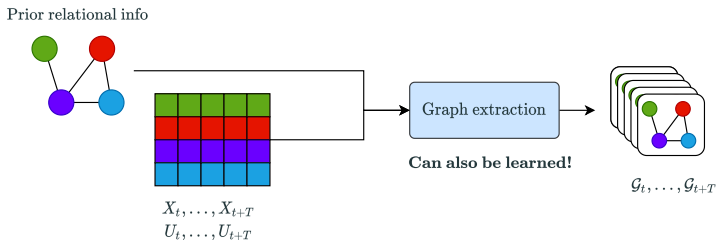
[5] Cini et al., "Filling the Gaps: Multivariate Time Series Imputation by Graph Neural Networks" ICLR 2022

[4] Cini et al., "Neural spatiotemporal forecasting" (in preparation)

Spatiotemporal graph signals (ii)

We can then model any multivariate time series $\mathbf{X}_{t:t+T} = \{\mathbf{X}_t, \dots, \mathbf{X}_{t+T}\}$, together with covariates and additional relational information as a sequence of attributed graphs

$$\mathcal{G}_{t:t+T} = \{\mathcal{G}_t, \dots, \mathcal{G}_{t+T}\}$$



Note that relational information ($\tilde{\mathbf{A}}_t$) **might not be known** a priori and we might be required to estimate the underlying graph directly from the data, taking **uncertainty** into account.

Spatiotemporal graph signals (iii)

Let us stop and think about the meaning of $\tilde{\mathbf{A}}_t$.

- In spatiotemporal graphs, we can interpret the adjacency matrix as representing **mutual constraints** on the time evolution of connected time series.
- Borrowing from relational DBs terminology, we might consider edges as (soft) **functional dependencies** among observed values at different time steps.

From this perspective, one could also look at spatiotemporal graph processing as a **regularization** of standard neural time series processing methods.

- Processing these data with a fully-connected net would require much more parameters.
- Think of convolutional neural networks processing subsequent frames in a video.
→ **Making use of spatial inductive biases is critical!**

One could consider downstream tasks in the context of spatiotemporal graphs. We focus on **node-level forecasting** for now, we will see node-level imputation in the second part.

- Given a **window** W of past observations, we want to predict **the next** H observations at each sensor, i.e., we want to estimate $\forall i$

$$\mathbf{x}_{t:t+H}^i \sim p(\mathbf{x}_{t:t+H}^i | \mathcal{G}_{t-W:t})$$

- For simplicity here we consider only **point forecasts**, i.e., we estimate $\hat{\mathbf{x}}_{t:t+H}^i$ s.t.

$$\hat{\mathbf{x}}_{t:t+H}^i \approx \mathbb{E} [p(\mathbf{x}_{t:t+H}^i | \mathcal{G}_{t-W:t})]$$

Since we are dealing with time series we have to make **some assumptions** on the underlying data generating process.

- We assume that the underlying process is **stationary**.
 - Model parameters are time independent.
- We also assume the graph to be **homogeneous**.
 - All sensors are of the same type.
 - This hypothesis can be easily relaxed.
- We assume to have a **known graph**.
 - You might try to relax this assumption in your course project.

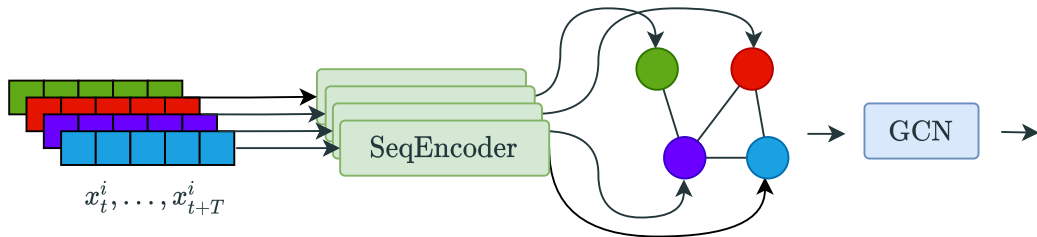
Spatiotemporal Graph Neural Networks

Any guess on how to process such data with graph neural networks?

The (simple) time-then-space approach (i)

A straightforward approach to process spatiotemporal graph is simply to:

1. **Embed** each node-level time series in a vector.
2. Use (a stack of) any of the **graph convolutional layers** we have seen so far.



The (simple) time-then-space approach (ii)

- Pros:**
- + Very simple, we can reuse the operators we already know.
 - + Theoretically is as expressive as it gets...
- Cons:**
- ...provided enough model capacity and depth.
 - Does not exploit spatiotemporal inductive biases.
 - Cannot handle changes in topology within the input window.

The idea is to implement gates and/or operators of architectures for sequential data by using graph convolutional layers. Or, conversely, implement message passing networks by using temporal operators.

We look at 3 different strategies, with increasing level of cohesion

- Interleaved spatial and temporal filters.
- Graph Convolutional Recurrent Neural Networks.
- Spatiotemporal Message Passing.

Spatiotemporal Graph Convolutional Networks (i)

We can build a deep spatiotemporal convolutional neural networks by **alternating spatial and temporal convolutional filters**.

The main idea:

- At layer l compute intermediate representations by using a node-wise temporal convolutional layer:

$$\mathbf{h}_{t-W:t}^{i,(l)} = \Theta^{1,l} \star_{\mathcal{T}} \mathbf{x}_{t-w:t}^{i,(l-1)}$$

where $\star_{\mathcal{T}}$ indicates the temporal convolution operator.

- Then, compute the updated representation by using a time-wise spatial convolution:

$$\mathbf{x}_t^{(l)} = \sigma \left(\tilde{\mathbf{A}}_t \mathbf{h}_t^{(l)} \Theta^{2,l} \right)$$

Spatiotemporal Graph Convolutional Networks (ii)

- Putting all together (with a slight abuse of notation) we get

$$\mathbf{x}_{t-W:t}^{(l)} = \sigma \left(\tilde{\mathbf{A}}_t \left(\Theta^{1,l} \star_{\mathcal{T}} \mathbf{x}_{t-W:t}^{(l-1)} \right) \Theta^{2,l} \right)$$

Clearly one can combine **any flavour of graph and temporal convolutions** to incorporate exogenous variables, edge attributes and so on.

It is also possible to substitute convolutional filters with self attention [6].

[6] Zheng et al., "Gman: A graph multi-attention network for traffic prediction", AAAI 2020

Examples

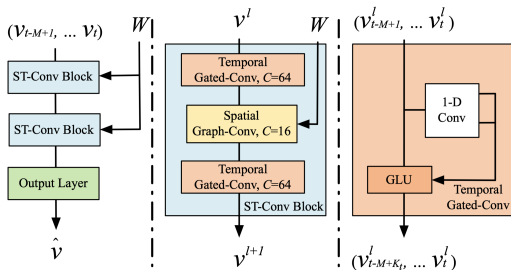


Figure 1: STGCN [7]

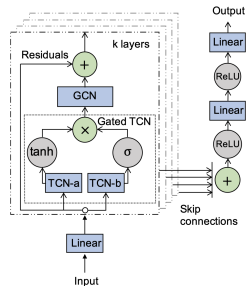


Figure 2: Graph Wavenet [8]

[7] Yu et al., "Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting", IJCAI 2018

[8] Wu et al., "Graph WaveNet for Deep Spatial-Temporal Graph Modeling", IJCAI 2019

Graph Convolutional Recurrent Neural Networks (i)

Let us consider a standard GRU cell.

$$\mathbf{r}_t^i = \sigma \left(\Theta_r [\mathbf{x}_t^i || \mathbf{h}_{t-1}^i] + \mathbf{b}_r \right) \quad (1)$$

$$\mathbf{u}_t^i = \sigma \left(\Theta_u [\mathbf{x}_t^i || \mathbf{h}_{t-1}^i] + \mathbf{b}_u \right) \quad (2)$$

$$\mathbf{c}_t^i = \tanh \left(\Theta_c [\mathbf{x}_t^i || \mathbf{r}_t^i \odot \mathbf{h}_{t-1}^i] + \mathbf{b}_c \right) \quad (3)$$

$$\mathbf{h}_t^i = (1 - \mathbf{u}_t^i) \odot \mathbf{c}_t^i + \mathbf{u}_t^i \odot \mathbf{h}_{t-1}^i \quad (4)$$

Note that here time series would be processed **independently** for each node (or alternatively as a single multivariate TS).

Any idea on how to integrate spatial convolutions?

Graph Convolutional Recurrent Neural Networks (ii)

We simply implement the gates by using spatial graph convolutions:

$$\mathbf{R}_t = \sigma \left(\tilde{\mathbf{A}}_t [\mathbf{X}_t || \mathbf{H}_{t-1}] \Theta_r + \mathbf{b}_r \right) \quad (5)$$

$$\mathbf{U}_t = \sigma \left(\tilde{\mathbf{A}}_t [\mathbf{X}_t || \mathbf{H}_{t-1}] \Theta_u + \mathbf{b}_u \right) \quad (6)$$

$$\mathbf{C}_t = \tanh \left(\tilde{\mathbf{A}}_t [\mathbf{X}_t || \mathbf{R}_t \odot \mathbf{H}_{t-1}] \Theta_c + \mathbf{b}_c \right) \quad (7)$$

$$\mathbf{H}_t = (1 - \mathbf{U}_t) \odot \mathbf{C}_t + \mathbf{U}_t \odot \mathbf{H}_{t-1} \quad (8)$$

Introduced in [9] and later popularized in the traffic forecasting context.

[9] Seo et al., "Structured Sequence Modeling with Graph Convolutional Recurrent Networks", ICONIP 2017

Graph Convolutional Recurrent Neural Networks (iii)

Or, if you prefer a different – more general – notation:

$$\mathbf{R}_t = \sigma(\text{GNN}_r(\mathcal{G}_t, \mathbf{H}_{t-1})) \quad (9)$$

$$\mathbf{U}_t = \sigma(\text{GNN}_u(\mathcal{G}_t, \mathbf{H}_{t-1})) \quad (10)$$

$$\mathbf{C}_t = \tanh(\text{GNN}_c(\mathcal{G}_t, \mathbf{R}_t \odot \mathbf{H}_{t-1})) \quad (11)$$

$$\mathbf{H}_t = (1 - \mathbf{U}_t) \odot \mathbf{C}_t + \mathbf{U}_t \odot \mathbf{H}_{t-1} \quad (12)$$

Again, the gates can be implemented by using any GNN, a popular choice in the literature is diffusion convolution [10]:

$$\mathbf{x}_t^{(l)} = \sum_{k=0}^K \left(\mathbf{D}_{t,0}^{-1} \tilde{\mathbf{A}}_t \right)^k \mathbf{x}_t^{(l-1)} \Theta^{1,k,(l)} + \left(\mathbf{D}_{t,l}^{-1} \tilde{\mathbf{A}}'_t \right)^k \mathbf{x}_t^{(l-1)} \Theta^{2,k,(l)} \quad (13)$$

[10] Li et al., "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting", ICLR 2018

Spatiotemporal Message Passing

A general scheme for spatiotemporal message-passing networks:

$$\mathbf{x}_{t-W:t}^{i,(l)} = \gamma \left(\mathbf{x}_{t-W:t}^{i,(l-1)}, \sum_{j \in \mathcal{N}(i)} \phi \left(\mathbf{x}_{t-W:t}^{i,(l-1)}, \mathbf{x}_{t-W:t}^{j,(l-1)}, \mathbf{e}_{t-W:t}^{ij,(l-1)} \right) \right)$$

We already saw this:

- ϕ is the message function.
- $\sum$ is a permutation invariant aggregation function (not necessarily the sum).
- γ is the update function.

The difference here is that instead of vectors we have sequences associated to node features.

→ We must use operators that work on sequences!

[4] Cini et al., "Neural Spatiotemporal Forecasting", (in preparation)

Example: Temporal Graph Convolution

We can use the above framework to design a temporal graph convolution as

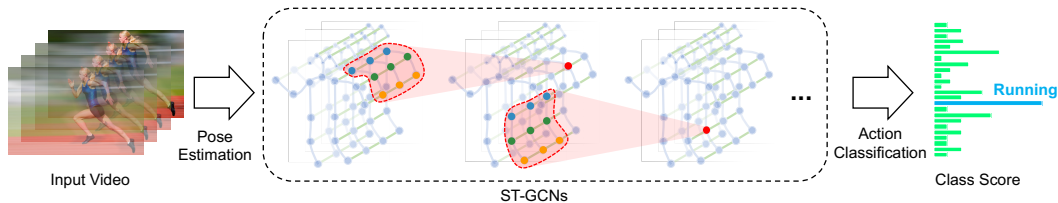
$$\mathbf{x}_{t-W:t}^{i,(l)} = \sigma \left(\Theta^{1,(l)} \star_{\mathcal{T}} \left[\mathbf{x}_{t-W:t}^{i,(l-1)} \parallel \sum_{j \in \mathcal{N}(i)} \left(\Theta^{2,(l)} \star_{\mathcal{T}} \left[\mathbf{x}_{t-W:t}^{i,(l-1)} \parallel \mathbf{x}_{t-W:t}^{j,(l-1)} \parallel \mathbf{e}_{t-W:t}^{ij,(l-1)} \right] \right) \right] \right),$$

or, more simply,

$$\mathbf{x}_{t-W:t}^{i,(l)} = \text{TCN}^{1,l} \left(\mathbf{x}_{t-W:t}^{i,(l-1)}, \sum_{j \in \mathcal{N}(i)} \text{TCN}^{2,l} \left(\mathbf{x}_{t-W:t}^{i,(l-1)}, \mathbf{x}_{t-W:t}^{j,(l-1)}, \mathbf{e}_{t-W:t}^{ij,(l-1)} \right) \right).$$

Product graph models

Some approaches in the literature take spatiotemporal data and arrange them in a single, big heterogenous graph (which is the **product graph** resulting from the temporal and spatial dependencies).



[11] Yan et al., "Spatial temporal graph convolutional networks for skeleton-based action recognition", AAAI 2018

Spatiotemporal imputation

The problem of missing data

So far, we assumed to deal with **complete sequences**

→ Valid observations for each location and time step

However, this is often not the case:

- Sensor networks are subject to faults – like readout failures or communication flaws.
 - The time series are affected by **missing data**

If we wish to use any of the methods presented before, we need a way to **impute**, i.e., reconstruct, missing observations.

Multivariate time series imputation

The problem of filling missing values in a (multivariate) sequence of data is often referred to as **multivariate time series imputation** (MTSI).



Let $\mathbf{X}_{t:t+T} = \{\mathbf{X}_t, \dots, \mathbf{X}_{t+T}\}$ be a multivariate time series with missing values. We group all valid observations into set $\mathcal{X}_{t:t+T} = \{\mathbf{x}_t^i \mid \mathbf{x}_t^i \in \mathbf{X}_{t:t+T}, \mathbf{x}_t^i \text{ is valid}\}$. We want to estimate **the missing observations**, i.e.,

$$\mathbf{x}_t^i \sim p(\mathbf{x}_t^i \mid \mathcal{X}_{t:t+T}) \quad \forall i, t \text{ such that } \mathbf{x}_t^i \notin \mathcal{X}_{t:t+T}$$

In principle, any forecasting method can be used for imputation, but we would not exploit **future observations** we have.

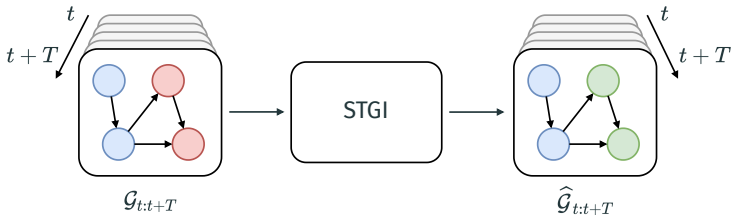
The common deep learning approach consists in using **autoregressive models** for sequential data (e.g., RNNs, TCNs).

Drawbacks:

- **relational information** (often strong in sensor networks) not taken into account.
- hard to capture **nonlinear time and space dependencies**.

Spatiotemporal graph imputation

Representing the input multivariate time series as a spatiotemporal graph signal, we can cast MTSI into **spatiotemporal graph imputation** (STGI).



We embed **relational constraints** – besides temporal ones – **explicitly** into the data processing:

$$\begin{array}{ccc} \mathbf{x}_t^i \sim p(\mathbf{x}_t^i | \mathcal{X}_{t:t+T}) & \mapsto & \mathbf{x}_t^i \sim p(\mathbf{x}_t^i | \mathcal{X}_{t:t+T}, \tilde{\mathbf{A}}_{t:t+T}) \\ \text{MTSI} & & \text{STGI} \end{array}$$

There are several methods that consider spatial information during processing (e.g., [12]).

However:

- Most of them account only for linear dependencies...
- ... or require prior physical knowledge on the processes involved.
- Often flexibility is limited.

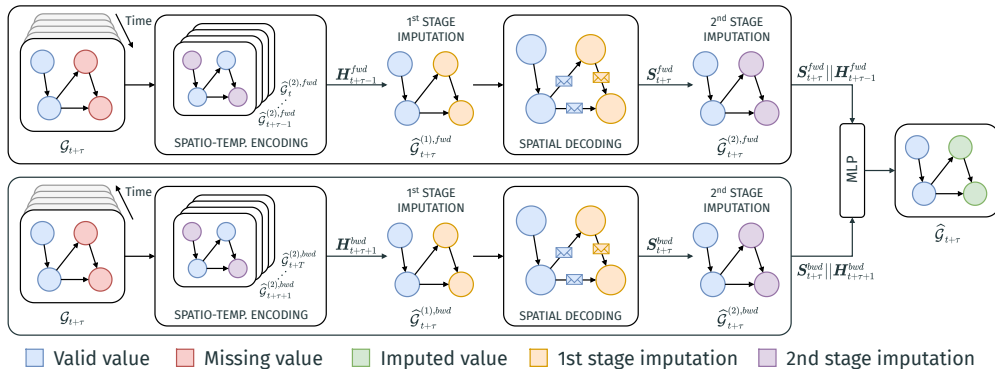
Graph-based methods – in particular, Graph Deep Learning – are appealing in this context:

- + High flexibility, given by working with graphs.
- Designing models that exploit all the available – useful – information is not trivial.

[12] Yi et al., "ST-MVL: Filling Missing Values in Geo-sensory Time Series Data", IJCAI 2016

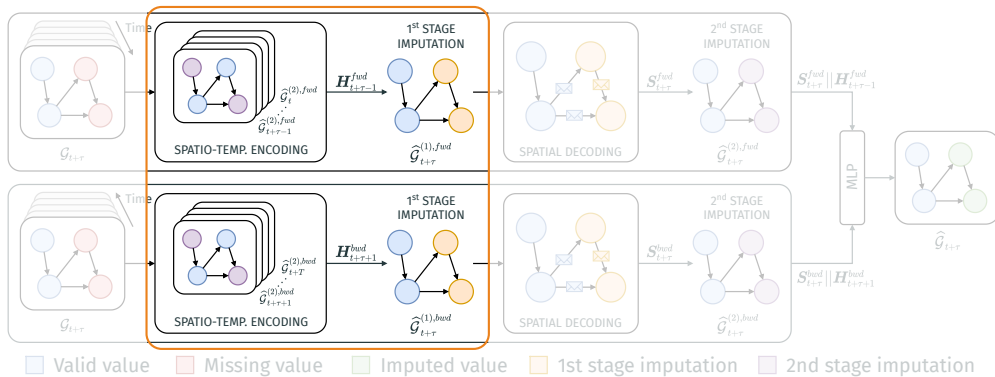
Graph Recurrent Imputation Network (GRIN)

GRIN is a graph-based, bidirectional, recurrent neural network which aims to reconstruct the input sequence by leveraging on both the temporal and spatial dimensions, jointly.



Graph Recurrent Imputation Network (GRIN)

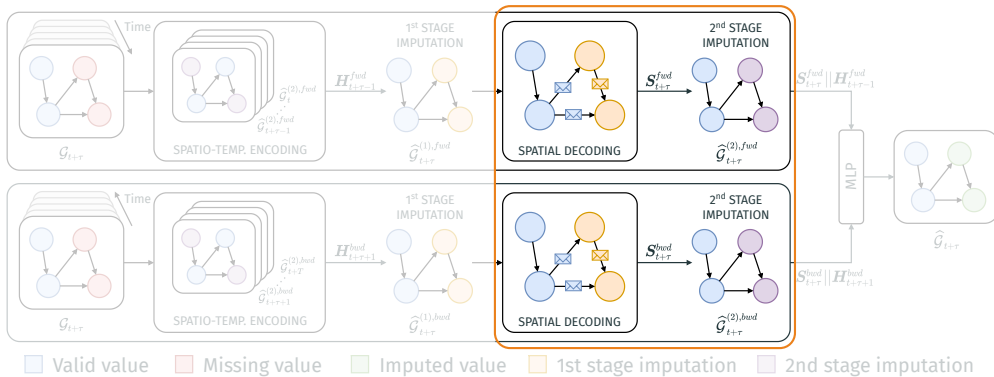
- 1 Feed a **recurrent GNN** with $\hat{\mathcal{G}}_{t+\tau-1}^{(2)}$ and obtain representation $\mathbf{H}_{t+\tau-1}$.
- 2 Impute missing features as one-step-ahead predictions from $\mathbf{H}_{t+\tau-1} \mapsto \hat{\mathcal{G}}_{t+\tau}^{(1)}$



Graph Recurrent Imputation Network (GRIN)

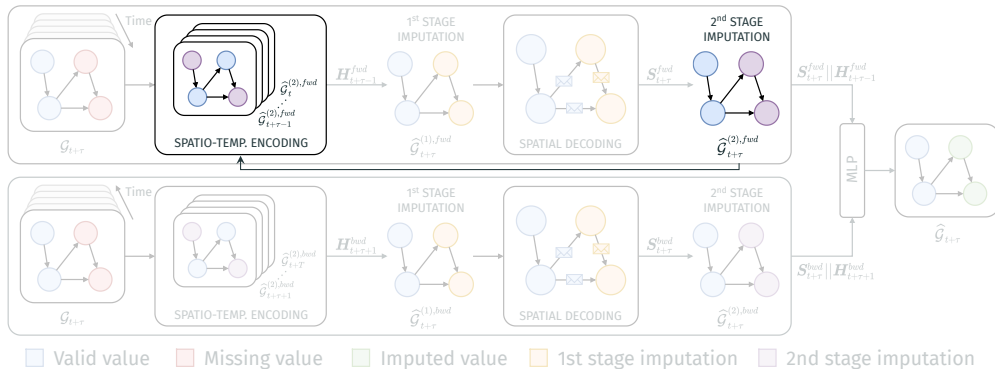
3 Exploit relationships between nodes at time $t + \tau$ through a **GNN** and obtain $\mathbf{S}_{t+\tau}$

4 Refine imputations using $\mathbf{S}_{t+\tau} \mapsto \hat{\mathcal{G}}_{t+\tau}^{(2)}$



Graph Recurrent Imputation Network (GRIN)

The 2nd stage imputation $\hat{\mathcal{G}}_{t+\tau}^{(2)}$ is then fed back to the recurrent GNN to update the state, obtaining representation $\mathbf{H}_{t+\tau}$.



Graph Recurrent Imputation Network (GRIN)

Obtain final imputations by combining (with an MLP) the **representations** extracted by processing the sequence in both forward and backward directions.

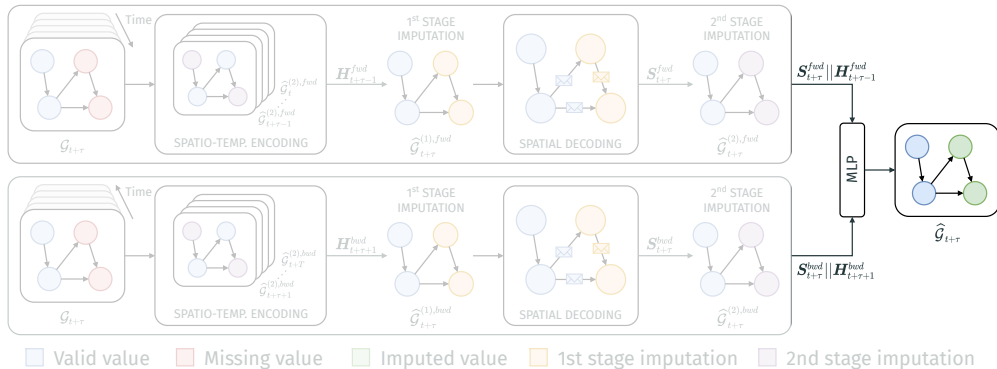


Table 1: Evaluation of different imputation methods on three real-world dataset: AQI (air quality monitoring), PEMS-BAY (traffic) and CER-E (smart grids). Performance averaged over 5 runs.

	AQI		PEMS-BAY		CER-E	
	MAE	MRE(%)	MAE	MRE(%)	MAE	MRE(%)
Mean	39.60 $\pm$ 0.00	59.25 $\pm$ 0.00	5.42 $\pm$ 0.00	8.67 $\pm$ 0.00	1.51 $\pm$ 0.00	71.51 $\pm$ 0.00
KNN	34.10 $\pm$ 0.00	51.02 $\pm$ 0.00	4.30 $\pm$ 0.00	6.88 $\pm$ 0.00	1.22 $\pm$ 0.00	57.71 $\pm$ 0.00
MF	26.74 $\pm$ 0.24	40.01 $\pm$ 0.35	3.29 $\pm$ 0.01	5.27 $\pm$ 0.02	1.01 $\pm$ 0.01	47.87 $\pm$ 0.36
MICE	26.98 $\pm$ 0.10	40.37 $\pm$ 0.15	3.09 $\pm$ 0.02	4.95 $\pm$ 0.02	0.98 $\pm$ 0.00	46.59 $\pm$ 0.23
VAR	22.95 $\pm$ 0.30	33.99 $\pm$ 0.44	1.30 $\pm$ 0.00	2.07 $\pm$ 0.01	0.53 $\pm$ 0.00	24.94 $\pm$ 0.02
rGAIN	21.78 $\pm$ 0.50	32.26 $\pm$ 0.75	1.88 $\pm$ 0.02	3.01 $\pm$ 0.04	0.71 $\pm$ 0.00	33.45 $\pm$ 0.16
BRITS	20.21 $\pm$ 0.22	29.94 $\pm$ 0.33	1.47 $\pm$ 0.00	2.36 $\pm$ 0.00	0.64 $\pm$ 0.00	30.07 $\pm$ 0.11
GCGRU	18.76 $\pm$ 0.11	27.79 $\pm$ 0.16	1.11 $\pm$ 0.00	1.77 $\pm$ 0.00	0.41 $\pm$ 0.00	19.51 $\pm$ 0.03
GRIN	14.73 $\pm$ 0.15	21.82 $\pm$ 0.23	0.67 $\pm$ 0.00	1.08 $\pm$ 0.00	0.29 $\pm$ 0.00	13.71 $\pm$ 0.03

GRIN estimates missing values at a node by using available information at **neighboring** nodes.

In the imputation task, we are interested in estimating missing values in **partially observed** sequences.

Question

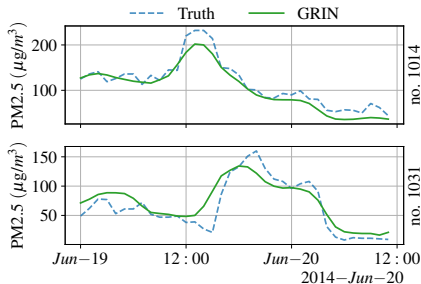
Can we use the same method to infer **observations** of **virtual sensors** at locations where we **do not** have physical ones?

Idea

Simulate the presence of a sensor by adding a node with **no data**, then let the model **infer** the corresponding time series.

Clearly, several assumptions are needed

- high-degree of homogeneity of sensors,
- capability to reconstruct from observations at neighboring sensors,
- ...



Coding Spatiotemporal GNNs



tsl (Torch Spatiotemporal) is a python library built upon PyTorch and PyG to accelerate research on neural spatiotemporal data processing methods, with a focus on **Graph Neural Networks**.

Demo

Summary

In this lecture we saw several methods to deal with inference problems on sets of time series by exploiting relational inductive biases.

Some takeaway points:

- In processing spatiotemporal data you have to deal with several subtleties to build a good model.
- Graph deep learning models are very flexible and can be extended to work in several different setting.
 - Even if it is not always trivial.
- When spatial dependencies exist use them, they will help a lot!

Questions?

References i

- [1] D. Zambon, C. Alippi, and L. Livi, "Concept drift and anomaly detection in graph streams," *IEEE transactions on neural networks and learning systems*, vol. 29, no. 11, pp. 5592–5605, 2018.
- [2] ———, "Change-point methods on a sequence of graphs," *IEEE Transactions on Signal Processing*, vol. 67, no. 24, pp. 6327–6341, 2019.
- [3] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, "Temporal graph networks for deep learning on dynamic graphs," *arXiv preprint arXiv:2006.10637*, 2020.
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